

TECHNICAL & COMMERCIAL PROPOSAL

185.4 kWp Rooftop Solar PV System

299 × 620 Wp solar panels | 2 × 80 kW grid-tied inverters

PREPARED FOR

Kampala Marriott Hotel

Mezzanine roof, Kampala, Uganda

PREPARED BY

NileSun Power Ltd

Contact: Mahil Fernando

Tel: +256 708669574 | +94 773 55 96 82

Email: sales@nilesunpower.com

Ref: NSP-261105-KMH | 05-Nov-2026

Valid until 05-Dec-2026 | Clean Energy - Brighter Tomorrow

185.4 kWp Grid-Tied Rooftop Solar PV System

Quote & Client Details

Quote No.	NSP-261105-KMH	Date	05-Nov-2026	Valid Until	05-Dec-2026
Client Name	Kampala Marriott Hotel	Contact No.	Mr. Jagath +256 791 811805	Client Email	
Site Address	Kampala Marriott Hotel, mezzanine roof, Kampala	Supply Type	Three-phase 400 V, UEDCL, main LV switchboard	VAT TIN No.	

185.4 kWp Solar array capacity	**160 kW** Inverters (2 × 80 kW)	**299 nos** 620 Wp solar panels	**USD 102,789** Turnkey price excl. VAT
797 kWh Avg. daily generation	**291.1 MWh** Est. annual generation	**USD 25,343** Savings in year 1	**4.0 years** Payback period

1. Executive Summary

NileSun Power Ltd is pleased to submit this proposal for the design, supply, installation and commissioning of a **185.4 kWp grid-tied rooftop solar PV system** on the mezzanine roof of the **Kampala Marriott Hotel**: **299 × 620 W N-type bifacial solar panels** on a ballasted aluminium structure at 10° tilt, with **two 80 kW three-phase string inverters** connected to the hotel's main LV switchboard.

The array generates about **291,051 kWh per year** (approx. 797 kWh/day). The hotel used 981,901 kWh in October 2026 (UEDCL bill), so every solar kWh is used on site and directly reduces grid purchases. At the UEDCL shoulder tariff of UGX 357/kWh (≈ USD 0.0871/kWh at UGX 4,100/USD) this saves about **USD 25,343 in the first year**. The turnkey price is **USD 102,789** (excl. VAT), paid back in about **4.0 years**, with an IRR of **25.9 %** (Sections 12 and 14).

Key Benefits

- Lower bills: solar energy covers part of the hotel's daytime load at USD 0.040/kWh versus USD 0.087/kWh from the grid.
- Fast payback (4.0 years) and USD 763,309 savings over 25 years.
- Long life: Astronergy N-type bifacial panels (15-year product, 30-year power warranty); remote monitoring of generation and savings.

2. Equipment Schedule (Bill of Materials)

#	Item	Specification	Qty	UoM
1	Solar panels	Astronergy ASTRO N7 CHSM66RN(DG)/F-BH 620 Wp N-type TOPCon bifacial, 2382 × 1134 × 30 mm (Annex 4)	299	pcs
2	String inverters	Sigenergy Sigen PV 80M1, 80 kW three-phase, 1100 V DC, with monitoring (Annex 3)	2	pcs
3	Mounting structure	Enerack ballasted aluminium system, 10° tilt, 6 × 2P + 1 × 1P tables, no roof penetration (Annex 5)	1	system
4	DC protection	DC isolator units (2-pole 1000 V 32 A) + DC SPD Type II – 1 each per string	16+16	pcs
5	AC combiner panel	MCCBs 2 × 160 A (inverter incomers), 400 A roof isolator, AC SPD Type II	1	set
6	Grid connection	Connection to the existing Masterpact NW10 H1 solar incomer on the main LV switchboard	1	set
7	Cabling	Solar cable 4 mm², AC 4C × 50 mm² and 4C × 240 mm² Cu XLPE/SWA, cable tray	1	lot
8	Monitoring & control	Smart data logger, energy meter, remote monitoring	1	set
9	Earthing	Array bonding, connection to building earth and lightning protection	1	set

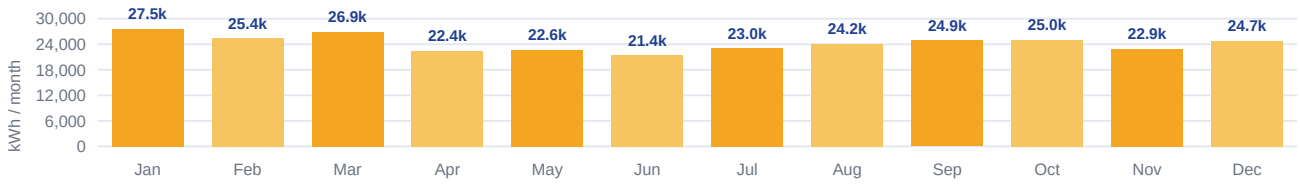
3. Solar Array & Inverters

Solar array	Value
Panels	299 × 620 W = 185.4 kWp
Arrangement	7 tables: 6 × 2P×23, 1 × 1P×23 (portrait)
DC / AC ratio	185.4 kWp / 160 kW = 1.16
Location / tilt	Flat concrete mezzanine roof, 10°
Panel area	approx. 808 m² (299 panels)

Inverters	Value
INV-01	152 panels = 94.2 kWp, 8 strings × 19
INV-02	147 panels = 91.1 kWp, 3 × 19 + 5 × 18
AC output	2 × 80 kW, 400 V 3N~
Connection	AC combiner → MLVS via riser

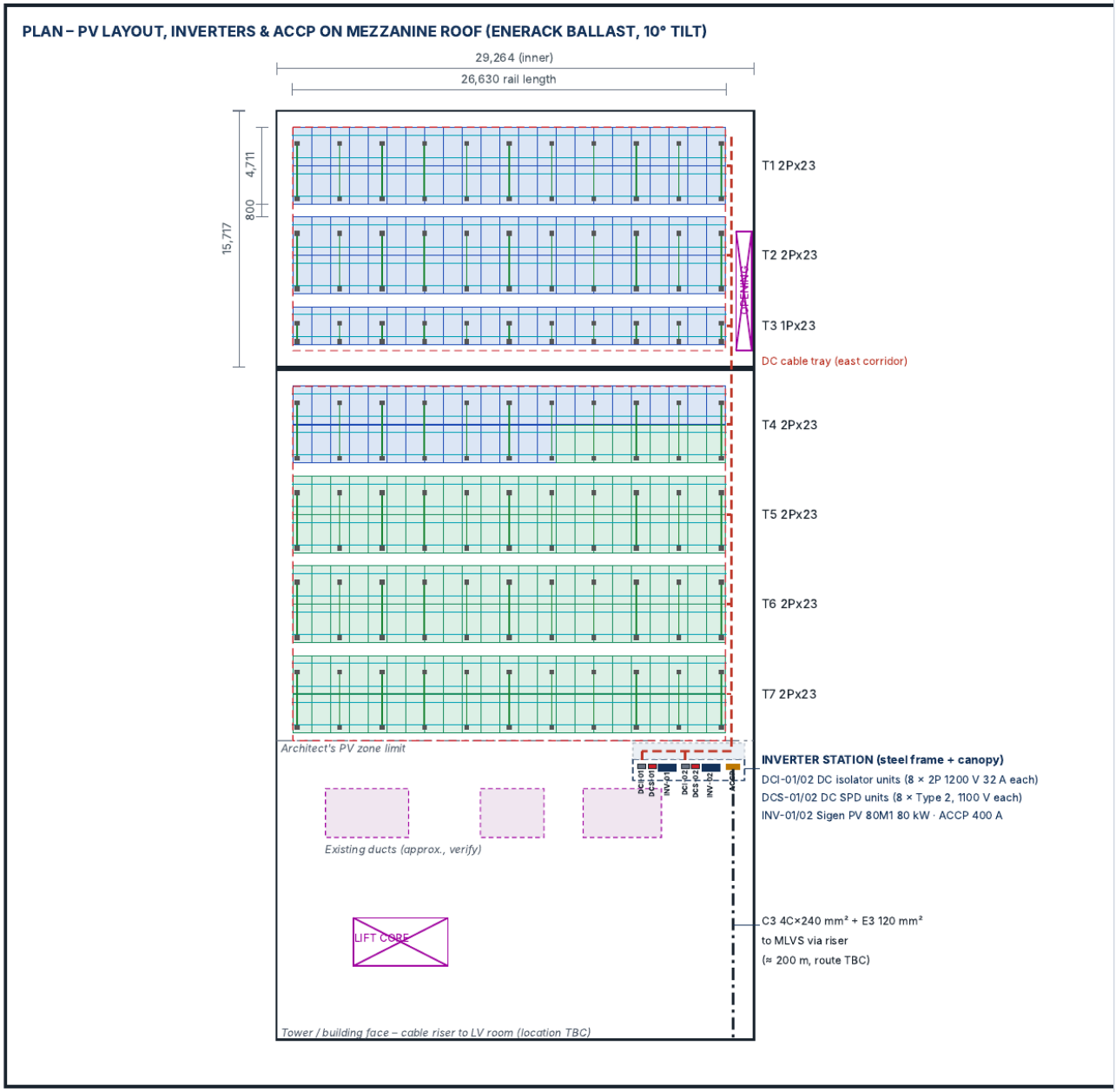
4. Estimated Energy Generation

PVsyst V8.1.8 simulation (P50, Meteonorm 9.0, PR 85.4 %, Annex 7). Annual total **291,051 kWh** (≈ 1,570 kWh/kWp).



5. Design Solar Top Installation (Roof Plan)

The 299 panels are placed on the mezzanine roof on 7 Enerack ballast tables at 10° tilt with 800 mm service aisles. INV-01 feeds tables T1–T4 (part), INV-02 feeds T4 (part)–T7. The inverter station and AC combiner panel sit at the south-east corner; the AC cable runs to the main LV switchboard via the building riser.



Proposed solar installation – Kampala Marriott Hotel, mezzanine roof (0°18'11.34"N, 32°35'13.76"E). Full drawing in Annex 2.

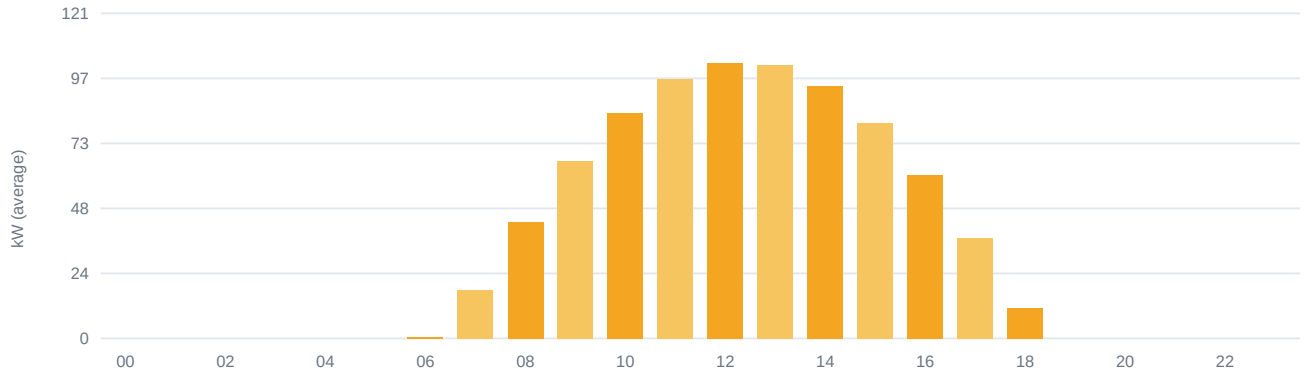
7. Hotel Load & Solar Contribution

The hotel used **981,901 kWh in October 2026** (UEDCL bill; ≈ 11.8 GWh per year), with an average daytime demand of about 1,348 kW and a maximum demand of 1,836 kW. The solar system supplies about **2.5 %** of this, so all solar output is consumed on site even on low-occupancy days – no export and no battery are needed.

Item	Value	Check	Result
Hotel annual consumption	11,782,812 kWh	Solar share of consumption	2.5 %
Solar generation, year 1	291,051 kWh	Peak solar output	≈ 103 kW (midday average)
Hotel daytime demand	far above solar output	Energy exported	0 kWh (all used on site)
Grid tariff (avoided)	USD 0.0871/kWh (UGX 357), +2 %/yr	Every solar kWh used?	Yes – offsets grid purchase

8. Solar Generation by Time-of-Use Band

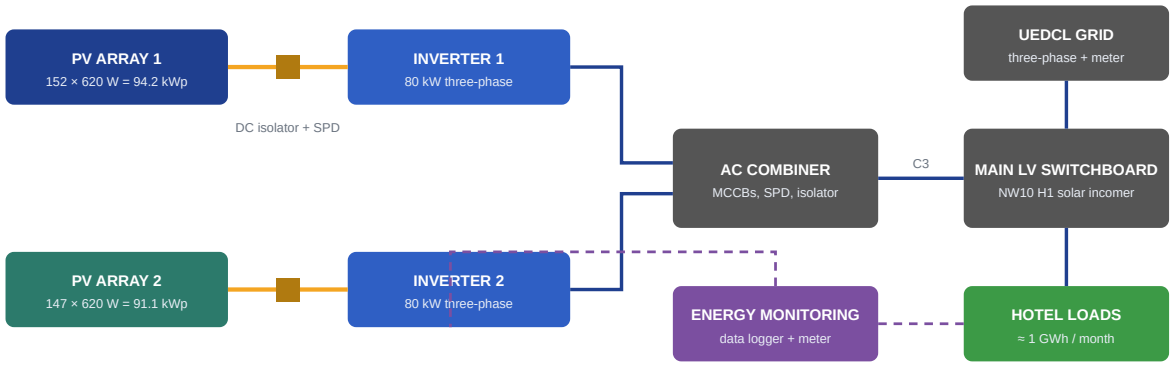
Average daily solar output by hour. All solar generation is counted in the shoulder period (06:00–18:00), when the hotel's air-conditioning, kitchen and laundry loads are high; no solar energy is assigned to the peak band.



TOU band	Hours	Hotel consumption (kWh/yr)	Solar energy (kWh/yr)	Share of solar	Reduction of band purchase
Off-peak	00:00–06:00	2,764,833	0	0.0 %	0.0 %
Shoulder	06:00–18:00	6,015,570	291,051	100.0 %	4.8 %
Peak	18:00–24:00	3,002,410	0	0.0 %	0.0 %
Total		**11,782,812**	**291,051**	**100 %**	**2.5 %**

*Consumption and tariffs from the UEDCL bill for October 2026 (tariff 30.2 Large Consumers); annual = October × 12. Rates are as billed (peak 428.8, shoulder 357.0, off-peak 286.5 UGX/kWh) at UGX 4,100/USD, excluding VAT; solar does not reduce the maximum-demand charge.

9. System Connection Diagram



10. Implementation Schedule

Phase	Activity	Duration
1. Site survey & design	Roof and LV switchboard survey, detailed design, approvals	Weeks 1–3
2. Procurement	Ordering, shipping (Mombasa), customs clearing, delivery to site	Weeks 2–10
3. Installation	Ballast structure, panels, inverters, AC combiner, cabling & earthing	Weeks 10–13
4. Commissioning	Testing, monitoring set-up, grid synchronisation	Weeks 13–14
5. Training & handover	Operator training, as-built drawings and O&M manual	Week 14

11. Warranty & After-Sales Support

Component	Warranty
Solar panels	15 yrs product / 30 yrs linear power
Inverters	5 years (extendable)
Mounting structure	10 years
Installation workmanship	2 years

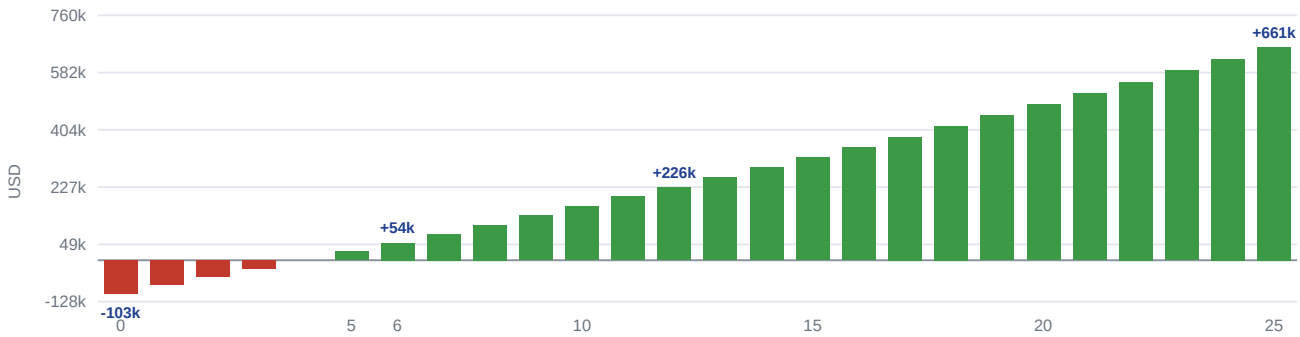
Installed to IEC solar installation standards and Uganda Electricity Regulatory Authority requirements. After handover we provide remote monitoring support, a free 6-month check-up visit and optional annual maintenance (panel cleaning, connection checks, inverter software updates, performance report).

12. Financial Analysis & Cash Flow

Assumption	Value
Turnkey investment (excl. VAT)	USD 102,789
Avoided tariff, year 1	USD 0.0871/kWh (shoulder)
Tariff escalation	2 % per year
Degradation	≈ 0.4 % per year
O&M cost / discount rate	none assumed / 10 %

Result	Value
Year-1 savings	USD 25,343
Simple payback	4.0 years
IRR (25 years)	25.9 %
NPV @ 10 %	USD 154,964
LCOE (solar energy cost)	USD 0.040/kWh
25-year savings	USD 763,309

Cumulative cash flow (USD) – the investment is recovered during year 4 (4.0 years).



Year	Generation (kWh)	Tariff (USD/kWh)	Savings (USD)	Net cash flow (USD)	Cumulative (USD)
0				(102,789)	(102,789)
1	291,051	0.087	25,343	25,343	(77,446)
3	285,840	0.091	25,895	25,895	(26,063)
5	283,558	0.094	26,726	26,726	26,970
6	282,424	0.096	27,151	27,151	54,121
7	281,294	0.098	27,583	27,583	81,704
10	277,932	0.104	28,922	28,922	167,116
12	275,713	0.108	29,850	29,850	226,349
15	272,418	0.115	31,298	31,298	318,780
20	267,013	0.127	33,870	33,870	482,906
25	261,715	0.140	36,654	36,654	660,520

No O&M or inverter-replacement cost is included. Values in brackets are negative (investment not yet recovered).

13. Quotation for 185.4 kWp Rooftop Solar PV – Turnkey EPC

Quote No.	NSP-261105-KMH	Date	05-Nov-2026	Valid Until	05-Dec-2026
Client	Kampala Marriott Hotel	Contact	Mahil Fernando	Currency	USD

#	Scope of supply & works	Qty	Unit	Price (USD)
1	PV modules – Astronergy ASTRO N7 620 Wp bifacial N-type TOPCon	299	nos	**USD 102,789**
2	String inverters – Sigenergy Sigen PV 80M1, 80 kW, 1100 V	2	nos	
3	Ballasted aluminium mounting structure, 10° tilt	1	lot	
4	DC system – PV cable, connectors, string isolators, cable tray	1	lot	
5	AC system – AC combiner panel, MCCBs, cabling to main LV switchboard	1	lot	
6	Smart data logger, energy meter & remote monitoring	1	set	
7	Earthing & lightning protection integration	1	lot	
8	Installation, testing & commissioning	1	lot	
9	Transport, logistics & customs clearing to site	1	lot	
	TOTAL TURNKEY EPC PRICE (excluding VAT and other applicable taxes)			**USD 102,789**

Equivalent to USD 0.55 per Wp. Amount in words: One hundred two thousand seven hundred eighty-nine US Dollars only.

Payment terms	75 % advance with PO · 25 % on commissioning
Delivery	Equipment 8–10 weeks from PO and advance; installation & commissioning ≈ 4 weeks after delivery
Taxes	VAT, withholding tax and statutory levies excluded – charged as applicable
Exclusions	Roof strengthening / waterproofing, battery storage, utility and permit fees, works not listed above
Client to provide	Roof access, connection to the existing NW10 H1 solar incomer on the main LV switchboard, power & water during installation

14. Financial Summary & Payback

USD 102,789 Total investment (excl. VAT)	**USD 25,343** Savings in year 1	**4.0 years** Payback period	**25.9 %** IRR (25 years)
USD 763,309 25-year savings	**USD 660,520** 25-year net gain after investment	**7.4 x** Savings / investment	**USD 154,964** NPV @ 10 %

15. Tariff Sensitivity

Avoided tariff (USD/kWh)	Year-1 savings (USD)	Payback (years)	IRR	NPV @ 10 % (USD)
0.0522	15,206	6.5	15.7 %	51,863
0.07	20,374	4.9	21.0 %	104,424
0.0871 (base)	**25,343**	**4.0**	**25.9 %**	**154,964**
0.10	29,105	3.5	29.6 %	193,230
0.1027	29,904	3.4	30.4 %	201,359
0.12	34,926	2.9	35.3 %	252,433

The project remains attractive over a wide tariff range. The break-even tariff (NPV = 0 at 10 %) is about **USD 0.0347/kWh** – roughly 60 % below today's shoulder tariff. Other rows: with the 40 % energy rebate = USD 0.0522; VAT not recoverable = USD 0.1027.

Annexes

Annex	Title	Page
Annex 1	Single-line diagram – PV system & grid connection	10
Annex 2	Solar PV layout – mezzanine roof	11
Annex 3	Data sheet – Sigenergy Sigen PV inverter (80 kW)	12–13
Annex 4	Data sheet – Astronergy ASTRO N7 620 W bifacial solar module	14–15
Annex 5	PV structural drawing – Enerack ballast system	16–17
Annex 6	Quotation – 185.4 kWp rooftop solar PV (one page)	18
Annex 7	PVsyst simulation report – P50 energy yield (V8.1.8, VC0)	19–25

Next Steps

1. Review of this proposal and clarification meeting with the hotel's engineering team.
2. Site survey: roof condition, LV switchboard, cable route and inverter location.
3. Confirmation of the tariff from recent utility bills and structural review of the roof.
4. Purchase order and advance payment – project start.

Thank you for considering NileSun Power Ltd. Contact: Mahil Fernando | +256 708669574 | sales@nilesunpower.com